

PLECTA: Overlap-aware instance reconstruction of dense filament networks from binary masks

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Abstract

Dense filament networks are difficult to partition into instances because crossings join distinct objects while segmentation gaps divide single objects. We present PLECTA, a deterministic method that converts a binary filament-axis mask into overlap-aware two-dimensional instances. PLECTA skeletonizes and decomposes the mask into arms and junctions, alternates exact per-junction continuation matching with globally gated gap matching, and re-estimates tangent, chord, and curvature evidence from the assembled chains. Crossing pixels may therefore belong to more than one output instance.

Evaluation used a method-independent partition of each input skeleton into common fragments. On 128 independently seeded held-out scenes from the same procedural generator, PLECTA achieved mean pairwise $F_1=0.854$ (density-stratified 95% bootstrap CI [0.845, 0.864]), precision 0.868, recall 0.843, recovery 0.799, and adjusted Rand index 0.851, with no failed scenes. Mean F_1 decreased from 0.911 at 20% coverage to 0.775 at 60%, identifying dense crossings as the main remaining regime of error. Development-only ablations showed that chord-turn evidence, gap bridging, and crossing matching made the largest contributions.

CNT-sheet scanning electron microscopy provides the application case. An optional upstream synthetic-data-driven U-Net supplies the binary mask; SEM width measurement and ribbon rendering are downstream characterisation stages and did not affect the held-out grouping result. The evidence supports same-generator synthetic generalisation and a development illustration on real SEM data, not general real-image or three-dimensional reconstruction claims.

Keywords: filament networks; instance segmentation; graph matching; geometric reconstruction; carbon nanotubes; scanning electron microscopy.

1 Introduction

The properties of carbon-nanotube (CNT) sheets, yarns, and composites depend on the geometry of their bundle networks: length, width, orientation, curvature, and contact topology influence mechanical and electrical response [1–5]. Scanning electron microscopy (SEM) resolves this projected network, but routine image analysis often stops at coverage or orientation distributions [6, 7]. Instance-resolved analysis instead requires each visible bundle to remain addressable across gaps, touches, and crossings.

This requirement is distinct from semantic segmentation. U-Net and its self-configuring variants can identify filament foreground [8, 9], but a binary foreground mask does not encode object identity. A faded segment splits one filament into several connected components; a crossing joins different filaments into one component. The ambiguity is greatest in dense networks, where several local continuations can be geometrically plausible. Compact-object instance

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priors—bounded blobs, centre-directed flows, or star-convex shapes—are poorly matched to long open curves that may overlap [10–12].

Existing methods address complementary parts of the problem. Ridge and vesselness filters enhance curvilinear signal but do not assign identity across junctions [13–15]. FIRE, CT-FIRE, SOAX, FiNTA, and DiameterJ extract fibres or network statistics in collagen, biopolymer, and fibrous materials [16–20]; methods such as that of Depriester et al. address fibre separation directly [21]. Learned filament-instance approaches include FibeR-CNN and recurrent pick-and-trace models [22, 23]. Orientation-aware branch separation followed by terminus pairing has also been used for microscopy filaments [24]. These approaches establish the importance of centreline and continuation evidence, but dense binary masks still require an explicit answer to which arms share an instance and which crossing pixels may belong to several instances.

CNT-specific workflows increasingly combine semantic segmentation with algorithmic tracing and transverse intensity profiles [25]. The present work focuses on the difficult middle step of that workflow: binary mask to overlap-aware instances. This separation is deliberate. It allows grouping to be evaluated independently of the upstream segmenter and prevents width rendering from changing an identity metric. An optional CycleGAN/U-Net route supplies masks for the SEM application [26, 27], but it is not the principal contribution.

We introduce PLECTA (Pairwise Linking with Evidence from Chain Tangents and Assignment), a deterministic graph-matching method for dense filament networks. PLECTA decomposes a skeleton into arms, endpoint stubs, and merged junction nodes. It then alternates exact per-node continuation matching with globally gated gap matching while tangent and curvature estimates expand from local arms to assembled chains. Its output is a set of projected 2-D instance layers in which crossing pixels may be shared. The name evokes the Latin *plectere*, to weave or interlace, without implying recovery of depth.

The study asks three questions. First, how accurately does the fixed method group method-independent skeleton fragments on unseen procedural scenes, including high-density cases? Second, which active geometric and topological components account for that performance, and how sensitive is it to mask degradation? Third, what does the method deliver in a CNT SEM workflow when the mask is derived manually or by a learned segmenter? Synthetic scenes provide exact overlap-aware identity; real SEM remains a development case. The intended claim is therefore a reproducible instance-reconstruction result under the tested generator, with explicit limits on distribution transfer, width validation, and 3-D interpretation.

2 Materials and methods

2.1 Scope and workflow

The formal input is a binary filament-axis mask $M \in \{0, 1\}^{H \times W}$. PLECTA reconstructs an overlap-aware collection $\{I_1, \dots, I_N\}$, where $I_n \subseteq M$ and a crossing pixel may belong to several instances. The evaluated grouping stage uses only M . A grayscale SEM image is optional downstream input for width and brightness measurement; it cannot revise the fixed grouping. Figure 1 separates this core task from upstream mask extraction and downstream rendering.

2.2 Synthetic data and evaluation split

Procedural scenes preserve the ordered centreline $\mathcal{C}_k$ and rendered support Ω_k of every filament k . Individual supports remain separate, so several filaments may own the same crossing pixel. The generator provides an undegraded one-pixel axis mask, a degraded thin mask containing small gaps and irregularities, a combined thick foreground, an SEM-like image, and the exact

Input-mask formation

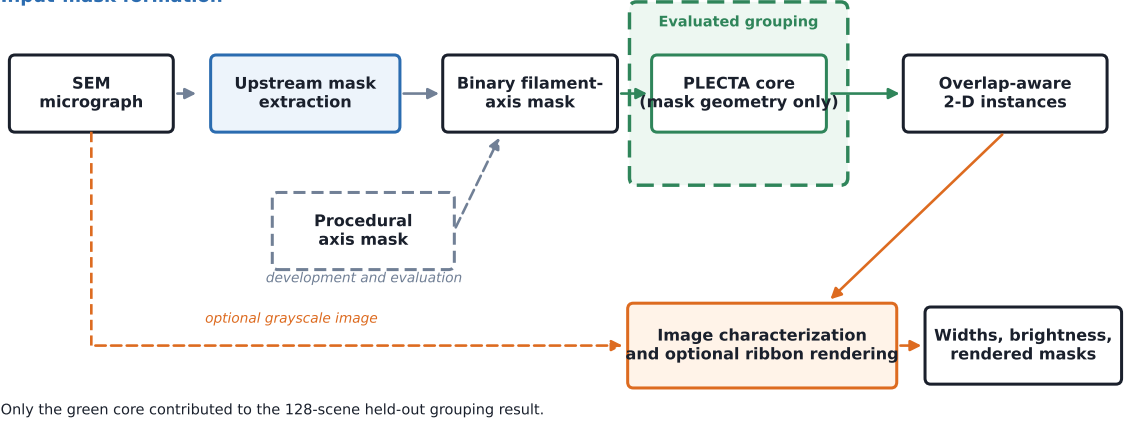


Figure 1. Workflow and evidence boundary. A micrograph is converted to a binary axis mask manually or by an upstream segmenter. PLECTA groups the mask into overlap-aware 2-D instances. SEM characterisation and ribbon rendering are optional downstream operations and were absent from the held-out grouping result.

overlap-aware instance reference (Figure 2). Areal coverage is the union area of the thick supports divided by image area; it is not the occupancy of the thin input mask.

Development used 84 scenes: 20 initial scenes across 20–60% coverage, 48 additional scenes at 20, 30, and 60%, and 16 scenes at 40%. The held-out set comprised 128 different seeds: 32 scenes at each of 20, 30, 40, and 60% coverage. It is an independent sample from the same generator, not an independent distribution. A separate 125-scene locked set remains unevaluated. The compact seed manifest records every seed block and an executable verifier checks counts, disjointness, and available scene metadata.

A separate controlled robustness set contained 50 scenes, ten at each of 20, 30, 40, 50, and 60% coverage. Each geometry was scored once from its clean axis and once from its degraded mask. This set was used only to quantify input-degradation sensitivity after the configuration was fixed.

2.3 CNT SEM data and mask sources

Two manually annotated CNT-sheet SEM fields were available for exploratory analysis: B58_100 (1024 × 1536 px, 280 filaments) and B58_110 (1024 × 1536 px, 430 filaments). The nominal horizontal field width is 1.66 μm; project metadata associate it with 1536 px, or approximately 1.08 nm px⁻¹. The annotation is one human interpretation of projected, locally ambiguous crossings. Both fields informed development and neither is a population-level test.

Two inputs were derived per field. The first was the skeleton of the manual instance annotation and is therefore an oracle-input condition, not an independent segmentation. The second was predicted by an upstream nnU-Net trained on paired synthetic images and masks; CycleGAN translation is one route from procedural masks to SEM-like training images [26, 27]. B58_100 appeared in that model’s training data, whereas B58_110 was verified absent from the relevant training set. Consequently, only B58_110 provides a training-clean observation of mask transfer, and $n = 2$ precludes inference.

Mask quality was compared after skeletonisation with a symmetric 3-px tolerance, avoiding a reward for mask thickness. Reconstruction scores were computed separately within each mask’s visible common-fragment population; because those populations differ, cross-mask F_1 values are

Thick-bundle / thin-axis synthetic scenes with exact ground truth

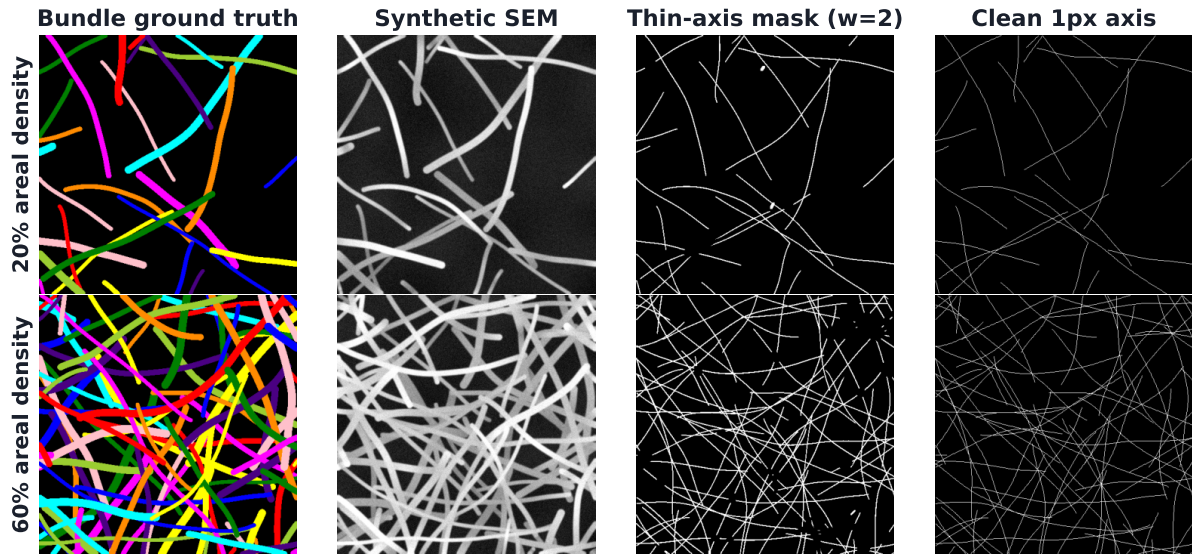


Figure 2. Procedural representations derived from the same filament geometry. Instance identity is retained through crossings; the degraded thin-axis mask is the primary algorithm input.

descriptive rather than paired estimates.

Author metadata required before submission: SEM instrument and detector, accelerating voltage, working distance, specimen preparation, CNT material identity, annotation protocol, and confirmation of the 1536-pixel calibration against the stored image dimensions. The upstream model archive must also supply its architecture, training split, and optimisation settings.

2.4 PLECTA reconstruction

2.4.1 Skeleton graph and local frames

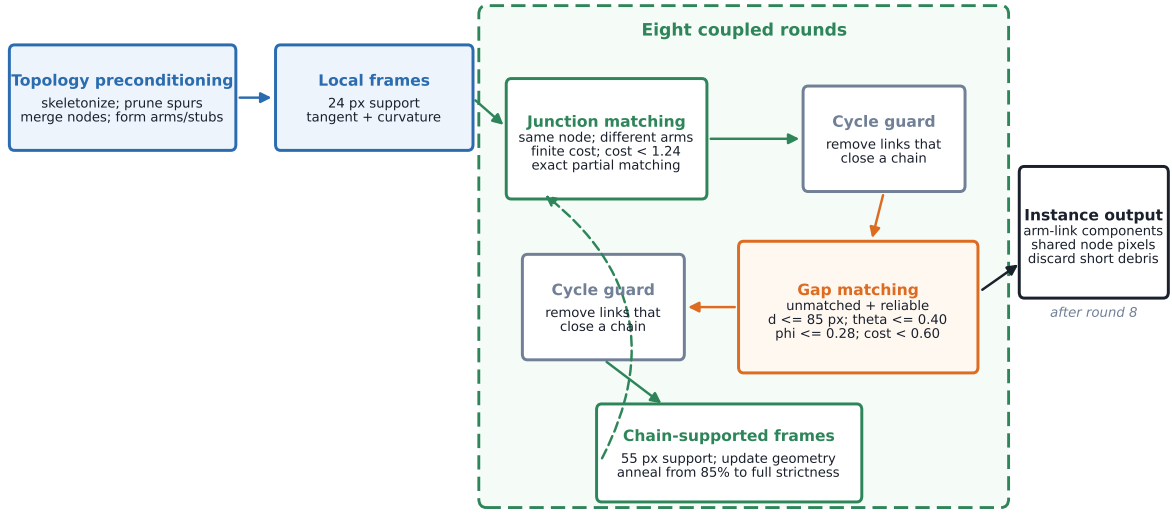
PLECTA binarises M at > 0 , skeletonises it, and iteratively removes spurs no longer than 3 px. Skeleton pixels with degree at least three form junction clusters, and adjacent clusters are merged into nodes. Connectors up to 5 px are absorbed into one node; connectors up to 14 px merge nearby nodes but remain real arms; free debris up to 2 px is absorbed. Each remaining connected path is stored as an ordered arm with one stub at each end.

A stub frame contains its tip, outward tangent, and signed curvature. These are estimated by an arclength polynomial fit and are considered reliable only with at least three samples spanning at least 3 px. Local frames use 24 px of support; after links form chains, the support expands to 55 px. Link cost combines direct tangent reversal, the turn from each tangent to the joining chord, and curvature continuity. Gap cost additionally penalises length. Short chords fade the chord-turn term rather than allowing an unstable angle to dominate.

2.4.2 Candidate gates and exact matching

Each of eight rounds first resolves junctions and then gaps (Figure 3). At a junction, candidates must share the node, use different arms, and have finite frames and cost. Exact maximum-weight partial matching selects disjoint pairs while allowing unmatched stubs. With the fixed unmatched price, a full-strictness junction edge is eligible only when its cost is below 1.24.

Gap candidates are constructed only from still-unmatched, reliable stubs. They must belong to



Gap links affect identity but are not painted into the core centreline output.

Figure 3. PLECTA stage and gate sequence. Topology preconditioning produces arms, stubs, and junction nodes. Every round applies exact per-junction matching, a cycle guard, global gated gap matching, a second cycle guard, and chain-supported frame re-estimation. The final chains are painted as overlap-aware instance layers.

different arms and different junctions, lie within 85 px, have direct tangent mismatch at most 0.40 rad, and have chord-turn mismatch at each end at most 0.28 rad. Global exact matching then accepts only edges whose cost is below 0.60 at full strictness. Cycles are broken after junction matching and again after gap matching. The unmatched prices and angular gap limits are annealed linearly from 85% to full strictness. Because only the eighth round is fully strict in the fixed schedule, that round is returned; there is no effective competition among several full-strictness rounds.

2.4.3 Instance output

Connected components of accepted arm links define instances. PLECTA paints every arm pixel and every junction cluster touched by a chain into that instance, so intersecting instances share crossing pixels. An isolated one-arm component shorter than 6 px is omitted. Accepted gap links determine identity but are not painted into the fixed centreline output. All effective parameters, including values previously inherited as code defaults, are materialised in the fixed JSON configuration.

2.5 Optional SEM characterisation and ribbon rendering

After grouping, perpendicular SEM profiles are sampled every 2 px along each ordered chain and at least 6 px from a junction. Profiles are recentred by at most 3 px and accepted only when contrast exceeds both four background-noise standard deviations and 0.05 normalised intensity; fitted widths must lie between 1.5 and 36 px. At least three valid cuts are required. These measurements attach apparent width, brightness, uncertainty, and a quality flag to an existing instance; they cannot alter its arm membership.

Optional ribbon rendering interpolates accepted gaps, resamples the centreline at 1 px, smooths it over 9 px with fixed endpoints, and draws the chain at its fitted width clipped to 3–40 px. It excludes junction blobs and does not clip the ribbon back to the input mask. Silhouette

Table 1. Held-out reconstruction scores overall and by target areal coverage. VI denotes variation of information; lower VI is better, whereas higher values are better for the other metrics.

Group	n	F_1	Precision	Recall	Recovery	ARI	VI split	VI merge
All	128	0.854	0.868	0.843	0.799	0.851	0.256	0.216
20%	32	0.911	0.928	0.896	0.918	0.907	0.137	0.093
30%	32	0.874	0.892	0.858	0.828	0.870	0.211	0.161
40%	32	0.858	0.864	0.853	0.796	0.855	0.242	0.224
60%	32	0.775	0.787	0.765	0.656	0.772	0.436	0.388

absorption would union rendered instances, but its threshold is zero by default. No absorption or later post-hoc continuation merge contributed to the 128-scene held-out result; neither is presented as part of the evaluated algorithm.

2.6 Metrics and statistical analysis

The primary endpoint is common-fragment pairwise F_1 . For each scene, the visible input skeleton is partitioned once into method-independent elementary fragments. Each fragment receives a procedural reference identity and a PLECTA instance identity. A true-positive pair shares both identities; a false-positive pair joins different reference filaments; and a false-negative pair separates fragments of one reference filament. Thus

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}. \quad (1)$$

This is a clustering metric related to the Rand family [28, 29]. It scores grouping, not rendered width, and weights filaments with many fragments more strongly because their pair count is quadratic. We therefore also report adjusted Rand index [30], variation of information split and merge components [31], and the fraction of reference filaments recovered by one dominant predicted instance using 0.8 purity and coverage thresholds.

Held-out means and density strata are accompanied by 95% nonparametric bootstrap intervals (20,000 replicates; seed 20,260,810). Aggregate resampling is stratified by density. The clean-minus-degraded analysis resamples paired scene differences within density. Active-component ablations are one-at-a-time development experiments on all 84 development scenes and are labelled as such; they do not reuse the held-out set.

3 Results

3.1 Held-out synthetic reconstruction

PLECTA completed all 128 held-out scenes with 0 failures. Mean common-fragment pairwise F_1 was 0.854 (density-stratified 95% bootstrap CI [0.845, 0.864]); mean precision, recall, reference-instance recovery, and adjusted Rand index were 0.868, 0.843, 0.799, and 0.851, respectively (Table 1). These scenes used unseen seeds but came from the same procedural generator as development data, so the result measures same-generator generalisation rather than distribution transfer.

Performance decreased with scene density (Figure 4a). Mean F_1 fell from 0.911 [0.888, 0.933] at 20% coverage to 0.775 [0.760, 0.790] at 60% coverage; the intermediate 30% and 40% means were 0.874 and 0.858. Recorded per-scene runtime had mean 2.54 s, median 1.12 s, 90th percentile 6.42 s, and maximum 34.06 s; these timings are hardware-specific.

Table 2. One-at-a-time component ablations on the 84-scene development set. Negative ΔF_1 values denote losses relative to the evaluated configuration.

Development variant	F_1	ΔF_1	ARI
Fixed configuration	0.867	+0.000	0.864
No Gap Bridging	0.728	-0.139	0.722
Single Round	0.847	-0.020	0.844
No Chain-Extended Frames	0.846	-0.021	0.842
No Annealing	0.863	-0.004	0.860
No Curvature Term	0.863	-0.005	0.859
No Join_Px	0.845	-0.022	0.841
No Crossing Merging At All	0.804	-0.063	0.799
Tangent Term Only	0.681	-0.186	0.675

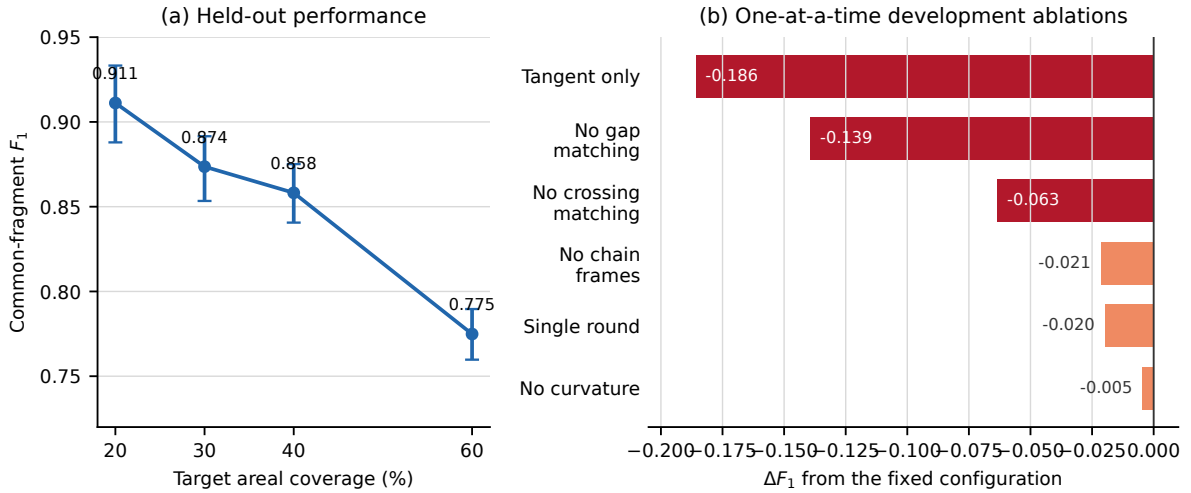


Figure 4. PLECTA performance across target areal coverage and component ablations. (a) Held-out mean common-fragment F_1 with 95% bootstrap intervals (32 scenes per level). (b) Development-only changes from selected one-at-a-time ablations; more negative values indicate larger losses.

3.2 Input-mask robustness

On the separate 50-scene controlled set, mean F_1 was 0.822 from degraded axes and 0.844 from the corresponding clean axes. The paired clean-minus-degraded change was 0.0224 (95% CI [0.0113, 0.0333]). Thus the modelled gaps and raster irregularities caused a small resolved loss under this generator, without testing robustness to a different image distribution.

3.3 Development ablations and downstream rendering

One-at-a-time ablations were evaluated only on the 84 development scenes (Table 2; Figure 4b). Relative to the evaluated configuration, using tangent evidence alone changed F_1 by -0.1858, removing gap matching by -0.1394, and removing crossing matching by -0.0634. Removing chain-extended frames and using a single round produced smaller changes of -0.0215 and -0.0199. These development results identify chord-aware continuation, gap matching, and crossing matching as the largest tested contributors; they are not held-out effect estimates.

Default Stage 4 ribbon rendering achieved development $F_1 = 0.860$, a change of -0.0068 from the core centreline output. The rendered instances had disconnected-component fraction 0.000 and branched fraction 0.005. Rendering therefore improved mask continuity while slightly changing the representation score, without changing chain membership. Enabling the separate

Table 3. Exploratory reconstruction from manual-derived and U-Net axes in two CNT SEM fields. Axis recall and precision compare the U-Net skeleton with the manual-derived skeleton using a symmetric 3-px tolerance.

Field	Input mask	Axis recall	Axis precision	F_1	ARI
B58_100	Manual-derived	—	—	0.882	0.882
B58_100	U-Net	0.782	0.753	0.438	0.435
B58_110	Manual-derived	—	—	0.837	0.836
B58_110	U-Net	0.729	0.887	0.452	0.449

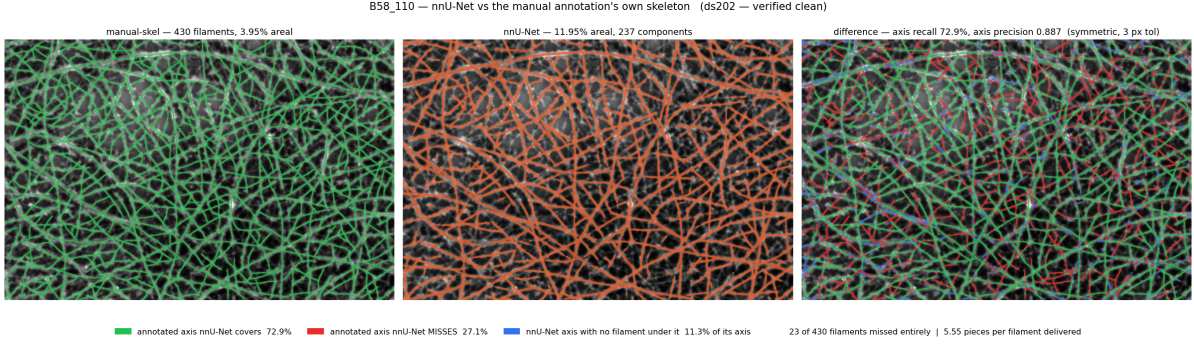


Figure 5. Training-clean B58_110 mask-source case. Left: the manual-instance-derived axis. Centre: the upstream U-Net axis. Right: tolerance-matched manual axis (green), missed manual axis (red), and U-Net axis without a matched manual axis (blue). The comparison illustrates upstream omissions and fragmentation; it is not a PLECTA output panel.

0.6 silhouette-absorption rule reduced F_1 further to 0.857. Because absorption can merge identities, it remained disabled and neither rendering condition contributed to the held-out grouping result. These topology measurements do not validate apparent width.

3.4 Two-field CNT SEM mask-source case

The two real SEM fields were exploratory development cases (Table 3). From the manual-derived axis, PLECTA reached $F_1=0.882$ on B58_100 and 0.837 on B58_110. These are oracle-input conditions because the axes were obtained from the manual instance annotations. With the upstream U-Net axes, the corresponding values were 0.438 and 0.452. The U-Net axis had recall/precision 0.782/0.753 on B58_100 and 0.729/0.887 on B58_110.

B58_100 was present in the segmenter’s training data; only B58_110 was verified training-clean. Moreover, each input mask defines a different visible common-fragment population, so the manual-derived and U-Net F_1 values are descriptive rather than paired. With two development fields, these observations show the practical dependence on upstream mask quality but do not estimate real-image generalisation.

4 Discussion

PLECTA addresses a narrow problem: assigning fragments in a binary filament-axis mask to overlapping projected instances. Its held-out common-fragment F_1 of 0.854 shows that explicit geometry can recover substantial identity without instance-labelled training data or SEM intensity. The mechanism is the combination of local and accumulated evidence. Exact matching at each junction selects mutually compatible continuations, gated gap matching reconnects separated arms, and repeated rounds replace short-arm direction estimates with tangent and curvature estimates from the chains assembled so far. Shared crossing pixels remain in each

applicable instance rather than being forced into one label.

The ablations support this interpretation but do not prove that every term is independently necessary. On the development set, using only tangent evidence reduced F_1 by -0.1858, removing gap bridging by -0.1394, and removing crossing matching by -0.0634. Preventing chain-extended frames produced a smaller decrease of -0.0215. Thus the largest gains arise from solving both kinds of discontinuity—junctions and missing mask segments—with complementary geometric evidence; iterative context provides an additional, more modest benefit. Because these tests reused development scenes, their values describe mechanism and configuration sensitivity, not independent effect estimates.

4.1 Density, masks, and optional rendering

Performance depends on the geometry preserved by the input mask. Mean held-out F_1 declined from 0.911 at 20% coverage to 0.775 at 60%, where crossings and locally plausible alternatives are more frequent. On 50 paired development scenes, the clean axis masks scored 0.0224 higher than degraded masks (95% CI 0.0113– 0.0333). PLECTA can bridge short omissions, but it cannot infer a filament absent from the mask or reliably reject a false axis introduced upstream. Supplying identical masks isolates grouping in an algorithmic comparison; it does not establish end-to-end SEM performance.

Image-informed ribbon rendering is downstream of grouping. On the development set it produced $F_1 = 0.860$, a change of -0.0068 relative to the core output, while yielding no disconnected instances and branch pixels in 0.005 of instances. This is a useful shape tradeoff rather than an improvement in identity. Silhouette absorption changes the instance partition and scored 0.857; it is therefore optional and should not be conflated with the fixed mask-only grouping result. Width and brightness can be reported for characterized instances, but their physical accuracy requires separate references.

4.2 Evidence boundary

The principal limitation is distributional. The 128 held-out scenes use independent seeds but the same procedural generator family used for development. Their confidence intervals quantify sampling variation within that family, not transfer to another simulator, specimen, microscope, or segmentation pipeline. The ablation and rendering studies are development-only and must remain secondary to the fixed held-out result.

The real SEM study is deliberately a case study with two fields, not a population validation. Manual-derived masks are an oracle-like input condition: they test grouping when the axis evidence is curated, not automatic SEM-to-instance performance. B58_100 is training-contaminated for the U-Net condition, whereas B58_110 is documented as training-clean; only the latter can illustrate an unseen upstream mask, and one field cannot estimate generalization. Moreover, manual-derived and U-Net masks generate different fragment universes. Their F_1 values therefore describe performance under each input condition but are not a paired comparison of identical objects. More independent fields, sealed before configuration, and annotations from multiple readers are required.

Two further scope limits concern the output itself. First, PLECTA recovers projected two-dimensional identity. Shared pixels encode overlap but not which filament passes above another, so no three-dimensional topology follows. Second, the grouping metric evaluates ownership of centreline fragments and is not a validation of rendered width. Quantitative diameter claims require matched physical measurements across the relevant scale, contrast, and specimen ranges. Pairwise fragment scores also emphasize instances containing many fragments; object recovery and ARI are retained as complementary views.

An independent procedural benchmark is a practical next test. The `merged_lines` mask generator in the `CNT_SEM` pipeline can export overlap-aware whole-bundle references and would add a different implementation and morphology model. It should supplement, not replace, the current generator: its field-of-view preset is presently calibrated to a narrow coverage regime, whereas the primary benchmark spans density. A multi-scene run should be made only after pooled calibration is fixed and its seeds are separated from any configuration work. No result from that generator is included in the present evidence. Broader validation should then extend to independent real masks and other filament domains while keeping input fragments and scoring rules fixed.

5 Conclusions

PLECTA reconstructs overlapping filament instances from a binary axis mask by alternating exact junction matching with gated gap matching as local arm geometry expands into chain-level evidence. The fixed method achieved common-fragment $F_1 = 0.854$ on 128 independently seeded held-out scenes from the same procedural generator, with performance declining as network density increased. Development ablations identify crossing matching, gap bridging, and tangent evidence as the principal contributors. Optional image-informed rendering improves geometric presentation but remains separate from the grouping claim. Two real SEM fields demonstrate the workflow under manual-derived and U-Net masks, not population generalization or width accuracy. The method therefore provides an inspectable, deterministic approach for projected filament identity; broader claims require independent real fields, validated physical widths, and an external-generator benchmark with a fixed calibration and sealed seeds.

CRediT author contribution statement

[To do: Fill in each author’s CRediT roles before submission, checked against actual contributions; keep the author names and order as listed above.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The PLECTA source, complete fixed configuration, compact seed manifest, and aggregate machine-readable evidence are available at <https://github.com/Jorallan/PLECTA>. Generated scenes, detailed per-scene records, and development diagnostics are retained locally. **[To do: Create a tagged archival release and add its DOI before submission; do not infer or preassign a DOI.]**

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